

SHIFT11

Microsoft-First Learning Resources

A deliberately narrow study path mapped to the revised nine-phase plan.

Learn each topic just before implementing it. Do not treat every link as a prerequisite for the MVP.

How to use this guide

Shift11’s core value is deterministic momentum detection, player attribution, traceable evidence and grounded explanation. Build the local analytics path before investing in advanced AI infrastructure.

Recommended sequence: Python/Git -> generator -> validation/tests -> momentum -> attribution/evidence -> Agent Framework + Foundry -> dashboard/integration.

Phase 1-3 | Foundation, generator, validation

Topic	Resource	Why it matters
Python fundamentals	https://learn.microsoft.com/en-us/training/paths/get-started-with-python-fundamentals/	Syntax, conditions, loops and functions.
Build Python programs	https://learn.microsoft.com/en-us/training/paths/get-started-with-python-programs/	Turn small functions into a runnable project.
Collections	https://learn.microsoft.com/en-us/training/modules/organize-data-with-collections-python/	Represent teams, players and events with lists/dictionaries.
File I/O	https://learn.microsoft.com/en-us/training/modules/read-write-files-python/	Save and load deterministic JSON output.
Reusable code	https://learn.microsoft.com/en-us/training/modules/write-reusable-code-python/	Separate models, generation, validation and analytics.
VS Code + Python	https://learn.microsoft.com/en-us/training/modules/install-configure-visual-studio-code-python/	Interpreter, environment and project execution.
Git fundamentals	https://learn.microsoft.com/en-us/training/modules/intro-to-git/	Track small changes and recover working versions.
GitHub Foundations	https://learn.microsoft.com/en-us/training/paths/github-foundations/	Repository branches, commits and project documentation.

Phase 4-5 | Momentum, attribution and evidence

No advanced machine learning is required for the first analytics engine. Focus on ordinary Python functions, clear metric definitions, rolling windows, event aggregation, test fixtures and readable output.

- Implement and test each momentum component independently before combining weighted scores.
- Create hand-authored event sequences that must cause a shift and sequences that must not cause one.
- Keep attribution deterministic and event-linked. Do not use an LLM to invent contribution scores.
- Use automated tests for threshold boundaries, persistence, cooldown, evidence ordering and missing data.

Phase 6 | Microsoft Agent Framework and Foundry

Resource	Use it for
Microsoft Agent Framework overview: https://learn.microsoft.com/en-us/agent-framework/overview/	Understand the Python-capable orchestration framework.
Agent Framework docs: https://learn.microsoft.com/en-us/agent-framework/	Look up only the APIs needed for the fact-pack explanation path.
Your first agent: https://learn.microsoft.com/en-us/agent-framework/get-started/your-first-agent?pivots=programming-language-python	Build a minimal Python agent once deterministic facts are ready.
What is Microsoft Foundry?: https://learn.microsoft.com/en-us/azure/foundry/what-is-foundry	Understand projects, models, tools and agent options.
Foundry SDKs and endpoints: https://learn.microsoft.com/en-us/azure/foundry/how-to/develop/sdk-overview	Choose the correct SDK/endpoint for the chosen integration.

AI integration gate: before connecting an agent, create one JSON fact pack and one deterministic explanation template. The AI should improve phrasing, not supply missing analytics.

Phase 7-9 | Demo, integration and submission

Resource	When / reason
Foundry Hosted Agents: https://learn.microsoft.com/en-us/agent-framework/hosting/foundry-hosted-agent	Phase 8 only if managed hosting improves the demo and account access is available.
Host Agent Framework in Foundry: https://learn.microsoft.com/en-us/azure/foundry/how-to/develop/framework-hosted-agents	Implementation guide for an existing agent that is ready to deploy.
Azure Container Apps: https://learn.microsoft.com/en-us/azure/container-apps/get-started	Optional alternative deployment path, not a prerequisite for core analytics.

Recommended learning order

1. Python basics, collections, functions and JSON file handling.
2. VS Code/environment basics and Git/GitHub.
3. Build the generator in Phase 2; study event semantics, formations, possession continuity and coordinates as needed.
4. Add validation and automated tests in Phase 3.
5. Learn only the Python needed for rolling windows, normalization, shift rules and contribution scoring.
6. Build and test the fact pack before starting Agent Framework/Foundry integration in Phase 6.
7. Build the minimum interactive demo; consider cloud hosting only after local execution is stable.
8. Finish end-to-end testing, documented demo steps and submission assets.

Intentionally out of scope until justified

Advanced ML training, RAG/vector databases, Kubernetes, complex MLOps, multi-agent orchestration and a relational database are not MVP prerequisites. Add complexity only when a demonstrated requirement makes it worthwhile.

Documentation note

Microsoft documentation changes over time. Recheck the linked pages when beginning Phase 6 or Phase 8, especially package prerequisites, SDK versions, hosting status and authentication requirements.